

Chapter 9 Fundamental sol. of the Laplace eq.

Let $\Omega \in \mathbb{R}^n$ be a domain, We call

$$-\Delta u(x) = 0, \quad x \in \Omega \quad (9.1)$$

A sol. (9.1) is called harmonic func.

§ 9.1 Fundamental sol. of the Laplace eq.

Let $n \geq 2$, $\Omega = \mathbb{R}^n$ and let $u(x) = v(|x|)$

be a radially symmetric func.

Find rad sym. sol. of (9.1)

Compute $(|x|^2)_{x_i}$

$$(|x|^2)_{x_i} \stackrel{\text{def of } |x|^2}{=} (x_1^2 + \dots + x_n^2)_{x_i} = 2x_i$$

$$(|x|^2)_{x_i} \stackrel{\text{chain rule}}{=} 2|x|(|x|)_{x_i}$$

chain rule.

$$\therefore 2x_i = 2|x|(|x|)_{x_i}$$

$$(|x|)_{x_i} = x_i / |x| \quad (x \neq 0).$$

Then by chain rule

$$(u(x))_{x_i} = (v(|x|))_{x_i}$$

Denote $r = |x|$.

$$\begin{aligned} \Rightarrow (v(|x|))_{x_i} &= (v(r))_{x_i} = \frac{dv}{dr}(r) \frac{dr}{dx_i} \\ &= \frac{x_i}{|x|} v_r(r) \end{aligned}$$

and

$$\begin{aligned} (u(x))_{x_i x_i} &= \left(\frac{x_i}{|x|} \cdot v_r(r) \right)_{x_i} \\ &= \frac{\delta_{ii}}{r} v_r(r) - \frac{x_i}{r^2} \cdot \frac{x_i}{r} v_r(r) \\ &\quad + \frac{x_i}{r} \cdot \frac{x_i}{r} v_{rr}(r) \end{aligned}$$

δ_{ii} is Kronecker's delta.

Therefore

$$\begin{aligned} \Delta u(x) &= \sum_{k=1}^n \left(u(x) \right)_{x_k x_k} \\ &= \sum_{k=1}^n \left(\frac{\delta_{kk}}{r} v_r(r) - \frac{x_k x_k}{r^3} v_r(r) + \frac{x_k x_k}{r^2} v_{rr}(r) \right) \\ &= \sum_{k=1}^n \left(\frac{1}{r} v_r(r) - \frac{x_k^2}{r^3} v_r(r) + \frac{x_k^2}{r^2} v_{rr}(r) \right) \\ &= \frac{n}{r} v_r(r) - \frac{1}{r} v_r(r) + v_{rr}(r) \\ &= \frac{n-1}{r} v_r(r) + v_{rr}(r). \end{aligned}$$

If u is harmonic func. then

$$v_{rr}(r) - \frac{n-1}{r} v_r(r) = 0.$$

Let $w := v_r$ then

$$w_r - \frac{n-1}{r} w = 0 \Leftrightarrow \frac{w_r}{w} = \frac{n-1}{r}$$

Separation Variable.

Integrating over $[1, r]$

$$\begin{aligned} \int_1^r \frac{w_r(\xi)}{w(\xi)} d\xi &= \int_1^r \frac{n-1}{r} d\xi \\ &= (n-1) \log r \end{aligned}$$

By the change of variable

$\eta = w(\xi)$ then $d\eta = w_r(\xi) d\xi$ and

$$\xi \parallel 1 \rightarrow r$$

$$\eta \parallel w(1) \rightarrow w(r)$$

$$\begin{aligned} \int_1^r \frac{w_r(\xi)}{w(\xi)} d\xi &= \int_{w(1)}^{w(r)} \frac{1}{\eta} d\eta \\ &= \log(w(r)) - \log(w(1)). \end{aligned}$$

$$\therefore \log(w(r)) - \log(w(1)) = (n-1) \log r$$

$$\Leftrightarrow \log(w_r(r)) = \log(w_r(1)) + (n-1) \log r$$

$$= \log \frac{|w_r(1)|}{r^{n-1}}$$

Assume $w > 0$ for any $r > 0$.
 $\Rightarrow \log w(r) = \log w(1) / r^{n-1}$

Recall $v_r = w$

Then for $r, \bar{r} > 0$.

$$v(r) - v(\bar{r}) = \int_{\bar{r}}^r w_r(s) ds$$

$$= \int_{\bar{r}}^r \frac{v_r(1)}{s^{n-1}} ds$$

$$= \begin{cases} v_r(1)(\log r - \log \bar{r}) & ; n=2 \\ -\frac{v_r(1)}{2-n} \left(\frac{1}{r^{n-2}} - \frac{1}{\bar{r}^{n-2}} \right) & ; n \geq 3 \end{cases}$$

Taking $r=1$ ($n=2$), $\bar{r} \Rightarrow \infty$ ($n \geq 3$).

$$\begin{cases} v_r(1)(\log r - \log \bar{r}) = C \log r \\ -\frac{v_r(1)}{2-n} \left(\frac{1}{r^{n-2}} - \frac{1}{\bar{r}^{n-2}} \right) \Rightarrow \frac{C}{r^{n-2}} \end{cases}$$

Def (Fundamental sol. of the Laplace eq)

For $x \in \mathbb{R}^n \setminus \{0\}$ define

$$\phi(x) = \begin{cases} -\frac{1}{2\pi} \log |x| & ; n=2 \\ \frac{1}{n(n-2)\alpha(n)} \frac{1}{|x|^{n-2}} & ; n \geq 3. \end{cases}$$

where $\alpha(n)$ is a volume of n -dim unit ball.
 We call ϕ a fundamental sol. the Laplace eq.

Computing the derivative ϕ

We find for $x \in \mathbb{R}^n \setminus \{0\}$

$$|\nabla \phi(x)| \leq \frac{C}{|x|^{n-1}}, \quad |\nabla^2 \phi(x)| \leq \frac{C}{|x|^n}$$

for some constant $C > 0$.

§ 9.2 Poisson eq. on whole sp.

Let ϕ be a fund sol. of the Laplace eq. defined by (9.2)

For $f \in C_0^\infty(\mathbb{R}^n)$

define

$$u(x) := (\phi * f)(x)$$

$$= \int_{\mathbb{R}^n} \phi(x-y) f(y) dy \quad (9.3)$$

$\phi * f$ is called Newtonian potential.

Thm 9.3

For $f \in C_0^\infty(\mathbb{R}^n)$ Let u be a Newtonian potential defined by (9.3). Then $u \in C_c^2(\mathbb{R}^n)$ and

$$-\Delta u(x) = f(x) \quad x \in \mathbb{R}^n$$

holds.

p.f. Fixed $x \in \mathbb{R}^n$

1. By the change of variable $z = x - y$.

$$u(x) = \int_{\mathbb{R}^n} \phi(x-y) f(y) dy$$

$$= \int_{\mathbb{R}^n} \phi(z) f(x-z) dz$$

Taking the derivative twice with respect to ∂^i

$$u_{x_i x_i}(x) = \int_{\mathbb{R}^n} \phi(z) f_{x_i x_i}(x-z) dz$$

$$u_{x_i x_i}(x) = \int_{\mathbb{R}^n} \phi(z) f_{x_i x_i}(x-z) dz$$

2. Since $\phi(z)$ is not defined at $z=0$,

we decompose

$$\begin{aligned} & \int_{\mathbb{R}^n} \phi(z) \Delta_x f(x-z) dz \\ &= \int_{B_\varepsilon(0)} \phi(z) \Delta_x f(x-z) dz \\ & \quad + \int_{\mathbb{R}^n \setminus B_\varepsilon(0)} \phi(z) \Delta_x f(x-z) dz \end{aligned}$$

$$= I_\varepsilon + J_\varepsilon$$

where $B_\varepsilon(0) = \{z \in \mathbb{R}^n; |z| < \varepsilon\}$
and $\varepsilon > 0$.

Since f is smooth.

$$\begin{aligned} |I_\varepsilon| &\leq \int_{B_\varepsilon(0)} |\phi(z) \Delta_x f(x-z)| dz \\ &\leq \sup_{y \in \mathbb{R}^n} |\Delta f(y)| \int_{B_\varepsilon(0)} |\phi(z)| dz \\ &\leq \sup_{y \in \mathbb{R}^n} |\Delta f(y)| \times \Gamma_A \end{aligned}$$

$$\Gamma_A = \begin{cases} C \varepsilon^2 \log \varepsilon & ; n=2 \\ C \varepsilon^2 & ; n \geq 3 \end{cases}$$

Note

$$\begin{aligned} & \int_{B_\varepsilon(0)} \log |z| dz \\ &= \int_0^\varepsilon r \log r dr \cdot 2\pi \\ &= 2\pi \left(\left[\frac{1}{2} r^2 \log r \right]_0^\varepsilon - \int_0^\varepsilon \frac{1}{2} r^2 \cdot \frac{1}{r} dr \right) ; n=2 \\ & \quad \downarrow \varepsilon^2 \log \varepsilon \quad \downarrow \varepsilon^2 \\ & \therefore J_\varepsilon \rightarrow 0 \end{aligned}$$

2. Let $R > 0$ be a const satisfying.

$$\text{supp}_z f(x-z) \subset B_R(0)$$

(we can take $R > 0$ since $f \in C^\infty(\mathbb{R}^n)$)

Then

$$\begin{aligned} J_\varepsilon &= \int_{\mathbb{R}^n \setminus B_\varepsilon(0)} \phi(z) \Delta_x f(x-z) dz \\ &= \int_{B_R(0) \setminus B_\varepsilon(0)} \phi(z) \Delta_x f(x-z) dz \end{aligned}$$

($\because \text{supp}_z f(x-z) \subset B_R(0)$).

$$\begin{aligned} &= \int_{B_R(0) \setminus B_\varepsilon(0)} \phi(z) \Delta_z f(x-z) dz \\ &= \left(\frac{\partial}{\partial x_i} f(x-z) = - \frac{\partial}{\partial z_i} f(x-z) \right) \\ &= - \int_{B_R(0) \setminus B_\varepsilon(0)} \nabla_z \phi(z) \cdot \nabla_z f(x-z) dz \\ & \quad + \int_{\partial B_R(0)} \phi(z) \nabla_z f(x-z) \cdot \nu(z) d\tau(z) \\ & \quad + \int_{\partial B_\varepsilon(0)} \phi(z) \nabla_z f(x-z) \cdot \nu(z) d\tau(z). \end{aligned}$$

$$\begin{aligned}
&= \int_{B_R(0) \setminus B_\varepsilon(0)} \underbrace{\Delta_2 \phi(z)}_{=0 \text{ } (\because \phi \text{ is fund. sol.})} f(x-z) dz \\
&\quad - \int_{\partial B_R(0)} \underbrace{f(x-z)}_{=0} \nabla_2 \phi(z) \cdot \nu(z) d\tau(z) \\
&\quad - \int_{\partial B_\varepsilon(0)} f(x-z) \nabla_2 \phi(z) \cdot \nu(z) d\tau(z) \quad K_\varepsilon \\
&\quad + \int_{\partial B_\varepsilon(0)} \phi(z) \nabla_2 f(x-z) \nu(z) d\tau(z) \quad L_\varepsilon \\
&=: -K_\varepsilon + L_\varepsilon
\end{aligned}$$

Estimate L_ε .

$$\begin{aligned}
|L_\varepsilon| &\leq \int_{\partial B_\varepsilon(0)} |\phi(z)| |\nabla_2 f(x-z)| d\tau(z) \\
&\leq \sup_{x \in \mathbb{R}^n} |\nabla f(x)| \left\{ \frac{\log |\varepsilon|}{\varepsilon^{2-n}} \times \int_{\partial B_\varepsilon(0)} d\tau(z) \times (\text{const.}) \right\} \\
&\leq C \begin{cases} \varepsilon \log |\varepsilon| & ; n=2 \\ \varepsilon & ; n \geq 3 \end{cases}
\end{aligned}$$

$$\rightarrow 0 \quad (\varepsilon \downarrow 0)$$

$$\therefore L_\varepsilon \rightarrow 0 \quad (\varepsilon \downarrow 0)$$

Compute K_ε .

$$\nabla_2 \phi(z) = \begin{cases} -\frac{1}{2\pi} \cdot \frac{1}{|z|} \cdot \frac{z}{|z|} & ; n=2 \\ \frac{1}{n(n-2)\alpha(n)} (2-n) |z|^{1-n} \frac{z}{|z|} & ; n \geq 3 \end{cases}$$

$$= -\frac{1}{n\alpha(n)} |z|^{-n} \cdot z$$

$$\nu(z) = -\frac{z}{|z|}$$

$$\begin{aligned}
-K_\varepsilon &= - \int_{\partial B_\varepsilon(0)} f(x-z) \nabla_2 \phi(z) \nu(z) d\tau(z) \\
&= -\frac{1}{n\alpha(n)} \int_{\partial B_\varepsilon(0)} f(x-z) \left(|z|^{-n} \cdot z \right) \cdot \left(|z|^{-1} \cdot z \right) d\tau(z)
\end{aligned}$$

$$= -\frac{1}{\underbrace{n\alpha(n)}_{\text{area of } \partial B_\varepsilon(0)} \varepsilon^{n-1}} \int_{\partial B_\varepsilon(0)} f(x-z) d\tau(z).$$

$$\rightarrow f(x) \quad (\varepsilon \downarrow 0)$$

$$\therefore u(x) = I_\varepsilon - K_\varepsilon + L_\varepsilon \rightarrow -f(x) \quad (\varepsilon \downarrow 0).$$